GPU scaling runs: ALIGNN-FF and VASP

B200 runs: NVIDIA B200 GPUs (SLURM partition b200, QOS blackwell_test).

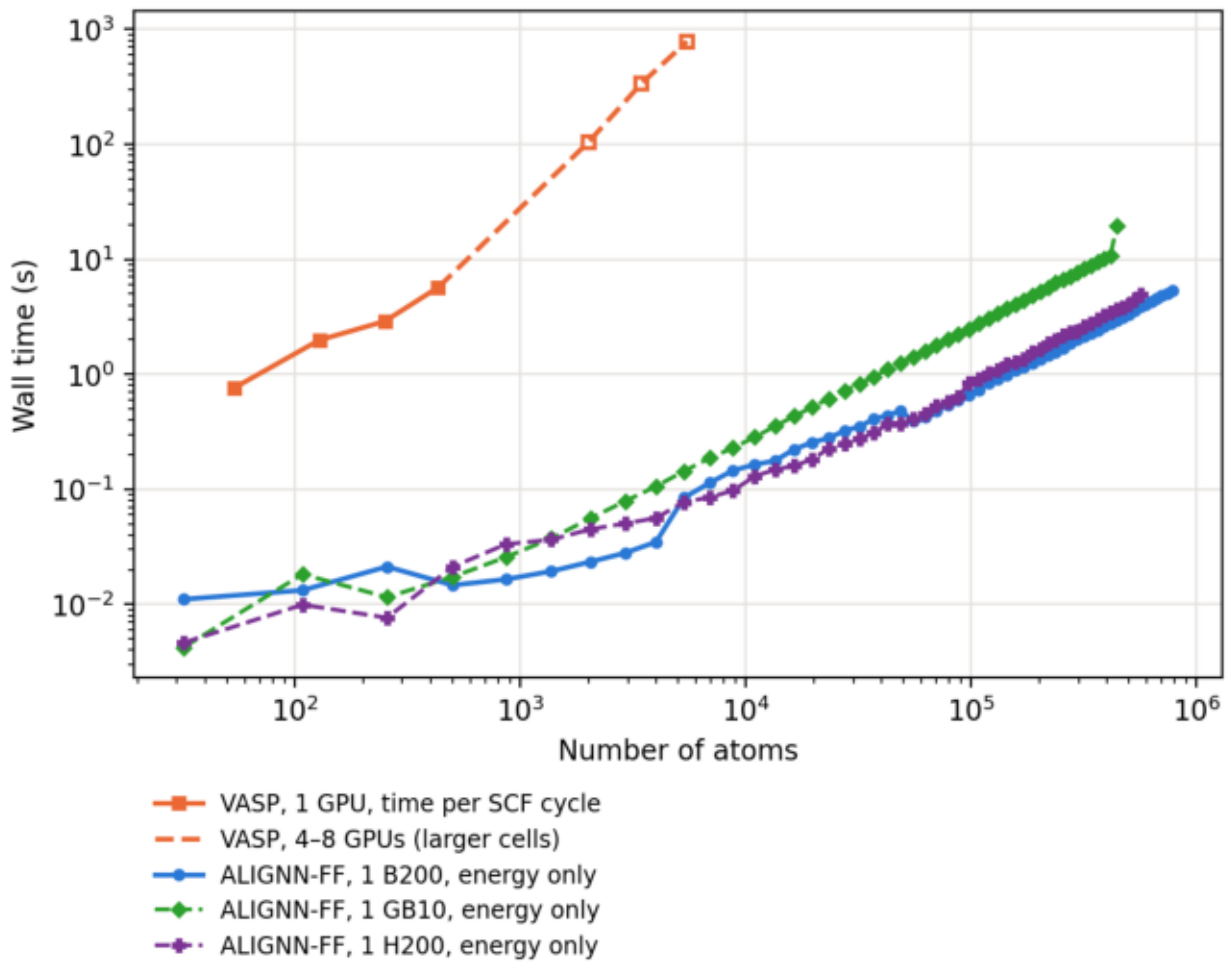
ALIGNN-FF, energy only (no forces), Cu FCC, 1 B200: 58 supercells, 1×1×1 to 58×58×58 (4 to 780,448 atoms).
Energies correct up to 442,368 atoms (48×48×48); larger sizes are affected by
the float32 drift and are shaded red in the table.

VASP, Si diamond: 21 runs over 9 supercell sizes (54 to 8,192 atoms); 19 completed,
2 crashed before the first SCF cycle (15×15×15 and 16×16×16).

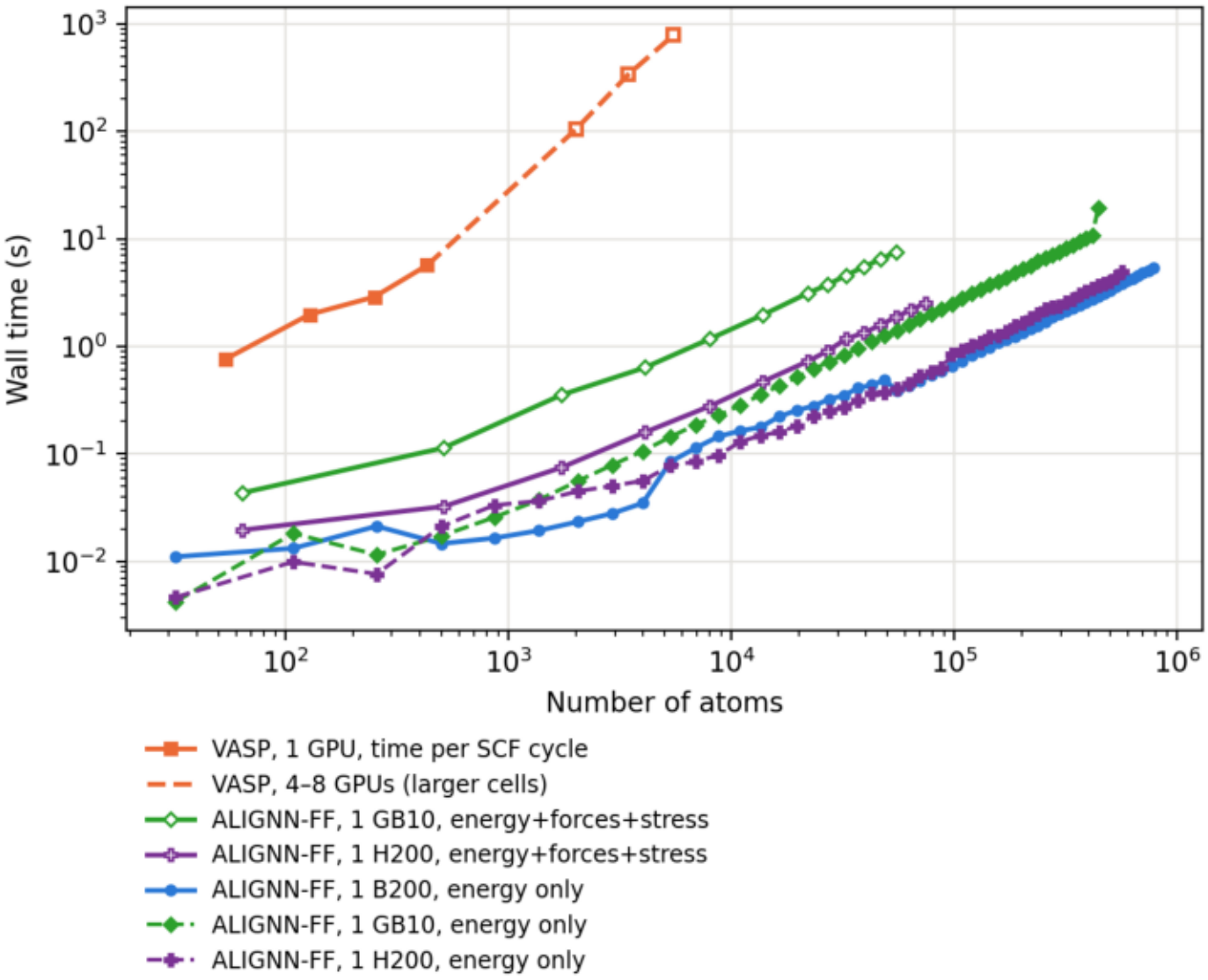
ALIGNN-FF, energy only, B200 settings rerun on a GB10 and an H200 (Cu FCC,
same checkpoint, pure-PyTorch model; energies match the B200 within 1e-5 eV/atom up to 500 atoms):
largest cell 442,368 atoms (GB10) and 562,432 (H200). At 256,000 atoms, inference takes
5.48 s (GB10), 0.70 s (H200) and 0.62 s (B200).

ALIGNN-FF with energy + forces + stress (added 2026-09-27; the MD workload), Si diamond, 1 GPU,
then-default matpes_r2scan checkpoint (hidden 128, 52 neighbors), same script on both cards:
GB10 (121.7 GiB): largest cell 54,872 atoms at 107 GB, OOM at 64,000;
H200 NVL (139.8 GiB): 74,088 atoms at 145 GB, OOM at 85,184.
Memory 1.95 MB/atom on both; the H200 is about 4× faster than the GB10 from ~4,000 atoms up.

Added 2026-11-02, GB10 only, same model (pages 13-14): 551,368 atoms energy only, 195,112 with forces
and layer checkpointing, and forces on 1B atoms chunked across 4 GPUs (1B added 2026-11-06).



All runs: VASP and ALIGNN-FF on B200, GB10 and H200



Energy only: filled markers (B200 run and its GB10/H200 reruns, Cu, 12 neighbors). Energy + forces + stress: hollow markers, solid lines (Si, 52-neighbor matpes_r2scan, the MD workload). VASP: time per SCF cycle, Si.

Which GPU for which job

	GB10	H200 NVL	B200
Energy only, 256,000 atoms: inference (s)	5.48	0.70	0.62
Energy only, 256,000 atoms: graph build (s, CPU)	1.12	1.50	1.07
Energy only, older FF on Cu: largest cell (atoms)	442,368	562,432	780,448
Energy + forces + stress, 21,952 atoms (s)	3.12	0.72	not run
E + F + S through ASE: largest cell (atoms)	54,872	74,088	not run
GPU memory	121.7 GiB (unified)	139.8 GiB	178.3 GiB

Reading the numbers
- Model evaluation: the H200 is within 13% of the B200 and 8x faster than the GB10 (energy only, identical settings and checkpoint). With forces and stress the H200 is 4.3x the GB10.
- Graph construction runs on the host CPU, so it follows the machine, not the GPU.
- For one evaluation, the largest cell is set by memory: 1.95 MB per atom with forces through ASE, 0.25 MB/atom energy only, the same on the GB10 and H200. Other paths and chunking go further (pages 13-14).

Recommendation
- Long training runs: GB10. Slowest per GPU, but locally available at no cost; multi-node training works (MPtrj trained on 8 GB10s).
- Fast turnaround: H200, ~4x the GB10 through ASE, but it is a shared, allocation-billed resource. Its 74k atoms is the largest ASE cell. The GB10 reaches 195k with checkpointing and 1B chunked on 4 GPUs.
- B200: fastest and largest on paper, but not available to this project; the H200 comes within 13% of it on model evaluation, so it is not worth pursuing.

Caveats: single-point evaluations, not full MD or training steps, so the ratios will shift somewhat for real workloads. Times with forces through ASE are three model passes.

Scaling to one billion atoms (with forces, through ASE)

GPU	Atoms per GPU	GPUs, no halo	Halo factor	GPUs with halo
GB10	54,872 (measured)	18,224	2.7x	48,702
H200 NVL	74,088 (measured)	13,497	2.5x	33,267
MI250X GCD (Frontier)	33,917 (memory rule)	29,484	3.1x	90,845 (= 11,356 nodes)

The governing number
- With forces the force field needs 1.95 MB of GPU memory per atom through ASE on both cards measured, so 10^9 atoms is ~2.0 PB before overheads. 'Memory rule' rows apply that figure, and the 96% of memory the H200 could use, to the card's memory; they are estimates, not measurements.
  A LAMMPS run would use forward_tensors (0.75 MB/atom measured), so these GPU counts overstate it.
- ALIGNN is a message-passing network, so when the crystal is split across GPUs each GPU also holds a ghost shell as thick as the receptive field (~20 A here: 2+2 layers x 5 A). Around a ~70k-atom Si block that multiplies the atoms each GPU holds by the halo factor. As-is, the GCD count exceeds Frontier (9,408 nodes).

Levers, largest first
- Evaluation path (measured 2026-11-01, same model): 0.195 MB/atom energy only, 0.75 with forces without the second-order graph, 0.53 with layer checkpointing. The second-order graph dominates the 1.95 MB, not the line graph.
- A leaner force field and half precision could cut memory further (unmeasured, and accuracy must be re-checked).
- Fewer message-passing layers shrink the halo: 20 A -> 10 A takes the factor from ~2.5x to ~1.6x.
- Memory per GPU matters more than speed when the whole cell is held at once. Chunking trades memory for time (measured to 1B atoms, pages 13-14).

Software
- A run this size needs LAMMPS domain decomposition with ghost-atom exchange. ALIGNN has a LAMMPS interface (pair_alignn, TorchScript); multi-layer ghost handling and an AMD build are not yet verified.
- The path is a ladder: 1 GPU -> 1 node -> 64 nodes -> thousands, fixing scaling at each step.

Time, if it fits
- At ~2.5 s per step per GPU (H200 at its largest cell, three model passes through ASE),
  1 ps at 1 fs steps is ~42 min of wall time
  plus communication; at thousands of nodes, every hour costs thousands of node-hours.

First steps (cheap, one GB10): memory per atom with forces for a leaner model (fewer neighbors, fewer layers, half precision), and pair_alignn on 2 GPUs with a real ghost region.

ALIGNN-FF results, energy only, B200 (1/2)

Supercell	Atoms	Neighbor list (s)	Graph (s)	Inference (s)	Total (s)	Model output (eV/atom)
1×1×1	4	0.0844	0.1568	3.2238	3.4650	0.604015
2×2×2	32	0.0015	0.0022	0.0073	0.0111	0.604014
3×3×3	108	0.0018	0.0026	0.0089	0.0132	0.604014
4×4×4	256	0.0087	0.0026	0.0100	0.0212	0.604014
5×5×5	500	0.0035	0.0029	0.0083	0.0147	0.604013
6×6×6	864	0.0047	0.0031	0.0086	0.0165	0.604013
7×7×7	1,372	0.0063	0.0035	0.0096	0.0194	0.604013
8×8×8	2,048	0.0085	0.0040	0.0109	0.0234	0.604013
9×9×9	2,916	0.0115	0.0046	0.0118	0.0279	0.604013
10×10×10	4,000	0.0158	0.0055	0.0136	0.0348	0.604013
11×11×11	5,324	0.0433	0.0181	0.0231	0.0845	0.604013
12×12×12	6,912	0.0680	0.0151	0.0304	0.1135	0.604013
13×13×13	8,788	0.0733	0.0248	0.0476	0.1457	0.604013
14×14×14	10,976	0.0881	0.0216	0.0542	0.1639	0.604013
15×15×15	13,500	0.1020	0.0252	0.0498	0.1771	0.604013
16×16×16	16,384	0.1218	0.0323	0.0682	0.2223	0.604013
17×17×17	19,652	0.1479	0.0399	0.0669	0.2547	0.604013
18×18×18	23,328	0.1598	0.0403	0.0786	0.2787	0.604013
19×19×19	27,436	0.1912	0.0387	0.0928	0.3226	0.604013
20×20×20	32,000	0.2106	0.0414	0.0985	0.3506	0.604013
21×21×21	37,044	0.2482	0.0462	0.1152	0.4096	0.604013
22×22×22	42,592	0.2590	0.0495	0.1283	0.4368	0.604013
23×23×23	48,668	0.2798	0.0548	0.1499	0.4845	0.604013
24×24×24	55,296	0.2096	0.0429	0.1373	0.3897	0.604013
25×25×25	62,500	0.2265	0.0481	0.1548	0.4294	0.604013
26×26×26	70,304	0.2547	0.0537	0.1738	0.4822	0.604015
27×27×27	78,732	0.2841	0.0601	0.1935	0.5377	0.604013
28×28×28	87,808	0.3112	0.0651	0.2156	0.5919	0.604013
29×29×29	97,556	0.3462	0.0750	0.2390	0.6602	0.604013
30×30×30	108,000	0.3773	0.0811	0.2643	0.7227	0.604013

ALIGNN-FF v6 (scale5b_v6.py), one B200, float32. Times are wall time for one energy evaluation.
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

ALIGNN-FF results, energy only, B200 (2/2)

Supercell	Atoms	Neighbor list (s)	Graph (s)	Inference (s)	Total (s)	Model output (eV/atom)
31×31×31	119,164	0.4516	0.0884	0.2913	0.8313	0.604013
32×32×32	131,072	0.4864	0.0953	0.3193	0.9011	0.604013
33×33×33	143,748	0.5267	0.1030	0.3502	0.9799	0.604013
34×34×34	157,216	0.5839	0.1139	0.3820	1.0798	0.604013
35×35×35	171,500	0.6101	0.1259	0.4164	1.1525	0.604013
36×36×36	186,624	0.6554	0.1350	0.4529	1.2434	0.604013
37×37×37	202,612	0.7091	0.1466	0.4912	1.3469	0.604013
38×38×38	219,488	0.7572	0.1677	0.5319	1.4568	0.604013
39×39×39	237,276	0.8124	0.1740	0.5753	1.5617	0.604013
40×40×40	256,000	0.8806	0.1872	0.6202	1.6880	0.604013
41×41×41	275,684	1.0001	0.1986	0.6675	1.8662	0.604013
42×42×42	296,352	1.0695	0.2132	0.7176	2.0003	0.604013
43×43×43	318,028	1.1327	0.2275	0.7691	2.1293	0.604015
44×44×44	340,736	1.1961	0.2483	0.8239	2.2683	0.604013
45×45×45	364,500	1.2776	0.2635	0.8810	2.4221	0.604013
46×46×46	389,344	1.3576	0.2852	0.9407	2.5835	0.604015
47×47×47	415,292	1.4514	0.3117	1.0032	2.7663	0.604015
48×48×48	442,368	1.5614	0.3171	1.0683	2.9467	0.604013
49×49×49	470,596	1.6539	0.3416	1.1366	3.1321	0.606979
50×50×50	500,000	1.7436	0.3695	1.2075	3.3206	0.639071
51×51×51	530,604	1.9523	0.3813	1.2820	3.6156	0.672090
52×52×52	562,432	2.0831	0.4097	1.3590	3.8518	0.702621
53×53×53	595,508	2.1964	0.4336	1.4393	4.0692	0.730894
54×54×54	629,856	2.3269	0.4530	1.5224	4.3024	0.757119
55×55×55	665,500	2.4602	0.4827	1.6091	4.5520	0.781473
56×56×56	702,464	2.5851	0.5138	1.6982	4.7971	0.804124
57×57×57	740,772	2.7414	0.5370	1.7917	5.0702	0.825215
58×58×58	780,448	2.9055	0.5709	1.9083	5.3847	0.844881

ALIGNN-FF v6 (scale5b_v6.py), one B200, float32. Times are wall time for one energy evaluation.
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

ALIGNN-FF results, energy only, GB10 (1/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
1×1×1	4	1.3637	0.0701	1.4338	0.0	0.604014
2×2×2	32	0.0017	0.0025	0.0042	0.0	0.604015
3×3×3	108	0.0145	0.0037	0.0182	0.1	0.604015
4×4×4	256	0.0050	0.0066	0.0115	0.1	0.604020
5×5×5	500	0.0053	0.0120	0.0173	0.2	0.604017
6×6×6	864	0.0068	0.0189	0.0257	0.3	0.604021
7×7×7	1,372	0.0087	0.0287	0.0374	0.4	0.604014
8×8×8	2,048	0.0122	0.0438	0.0560	0.6	0.604022
9×9×9	2,916	0.0148	0.0639	0.0787	0.8	0.604024
10×10×10	4,000	0.0195	0.0865	0.1060	1.0	0.604012
11×11×11	5,324	0.0279	0.1159	0.1437	1.4	0.604015
12×12×12	6,912	0.0326	0.1538	0.1864	1.8	0.604044
13×13×13	8,788	0.0367	0.1924	0.2291	2.3	0.604008
14×14×14	10,976	0.0458	0.2384	0.2841	2.8	0.603933
15×15×15	13,500	0.0608	0.2931	0.3539	3.5	0.603850
16×16×16	16,384	0.0770	0.3540	0.4310	4.2	0.603873
17×17×17	19,652	0.0910	0.4241	0.5151	5.0	0.603919
18×18×18	23,328	0.1104	0.5035	0.6139	5.9	0.603956
19×19×19	27,436	0.1265	0.5906	0.7171	7.0	0.603956
20×20×20	32,000	0.1427	0.6841	0.8268	8.1	0.603860
21×21×21	37,044	0.1612	0.7881	0.9493	9.4	0.603743
22×22×22	42,592	0.1921	0.9065	1.0986	10.8	0.603745
23×23×23	48,668	0.2169	1.0358	1.2527	12.4	0.603927
24×24×24	55,296	0.2349	1.1735	1.4083	14.0	0.604198
25×25×25	62,500	0.2678	1.3306	1.5985	15.9	0.604642
26×26×26	70,304	0.3060	1.4957	1.8016	17.8	0.605064
27×27×27	78,732	0.3374	1.6727	2.0101	20.0	0.605457
28×28×28	87,808	0.3540	1.8783	2.2323	22.3	0.605647
29×29×29	97,556	0.4018	2.0760	2.4778	24.8	0.605734
30×30×30	108,000	0.4980	2.2980	2.7959	27.4	0.605611

B200 settings rerun on one NVIDIA GB10: Cu FCC, 5 A cutoff, 12 neighbors, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (matches the B200 DGL energies within 1e-5 eV/atom up to 500 atoms). One call per size; the first row i GPU warm-up. Largest size shown is the largest that completed; the next (49×49×49 = 470,596 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600).
Timings are unaffected. atomgptlab job 13484, 2026-09-28.

ALIGNN-FF results, energy only, GB10 (2/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
31×31×31	119,164	0.5406	2.5462	3.0868	30.2	0.604993
32×32×32	131,072	0.5858	2.7901	3.3759	33.2	0.604449
33×33×33	143,748	0.6294	3.1341	3.7635	36.5	0.603988
34×34×34	157,216	0.6864	3.3575	4.0439	39.9	0.603673
35×35×35	171,500	0.7332	3.6670	4.4002	43.5	0.602742
36×36×36	186,624	0.8208	3.9880	4.8089	47.3	0.602057
37×37×37	202,612	0.8541	4.4270	5.2811	51.4	0.601320
38×38×38	219,488	0.9349	4.6847	5.6196	55.6	0.601225
39×39×39	237,276	1.0287	5.1840	6.2126	60.1	0.600929
40×40×40	256,000	1.1166	5.4838	6.6004	64.9	0.600853
41×41×41	275,684	1.1886	5.8977	7.0864	69.9	0.600795
42×42×42	296,352	1.2488	6.3326	7.5814	75.1	0.600620
43×43×43	318,028	1.3689	6.8675	8.2364	80.6	0.600555
44×44×44	340,736	1.4533	7.3131	8.7664	86.4	0.600509
45×45×45	364,500	1.4925	8.0306	9.5232	92.4	0.600274
46×46×46	389,344	1.7065	8.4010	10.1075	98.7	0.600081
47×47×47	415,292	1.8319	8.9564	10.7883	105.2	0.599898
48×48×48	442,368	1.9177	17.4482	19.3659	112.1	0.600148

B200 settings rerun on one NVIDIA GB10: Cu FCC, 5 A cutoff, 12 neighbors, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (matches the B200 DGL energies within 1e-5 eV/atom up to 500 atoms). One call per size; the first row i GPU warm-up. Largest size shown is the largest that completed; the next (49×49×49 = 470,596 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600).
Timings are unaffected. atomgptlab job 13484, 2026-09-28.

ALIGNN-FF results with forces, GB10

Supercell	Atoms	Time (s)	Peak mem (GB)	MB/atom	Energy (eV/atom)	Max |F| (eV/A)
2×2×2	64	0.043	0.2	3.03	-8.7476	2.0e-05
4×4×4	512	0.114	1.1	2.09	-8.7476	3.5e-05
6×6×6	1,728	0.353	3.4	1.99	-8.7476	3.8e-05
8×8×8	4,096	0.635	8.1	1.97	-8.7475	3.8e-05
10×10×10	8,000	1.174	15.7	1.96	-8.7479	3.9e-05
12×12×12	13,824	1.959	27.0	1.96	-8.7479	3.9e-05
14×14×14	21,952	3.116	42.9	1.95	-8.7476	5.6e-05
15×15×15	27,000	3.742	52.7	1.95	-8.7470	5.7e-05
16×16×16	32,768	4.576	64.0	1.95	-8.7464	5.8e-05
17×17×17	39,304	5.504	76.8	1.95	-8.7467	5.7e-05
18×18×18	46,656	6.454	91.1	1.95	-8.7470	5.7e-05
19×19×19	54,872	7.592	107.1	1.95	-8.7475	5.7e-05
20×20×20	64,000	OOM	–	–	–	–

Energy + forces + stress single point on one NVIDIA GB10: Si diamond (8-atom conventional cell, i×i×i), then-default matpes_r2scan checkpoint (hidden 128, 2+2 layers, 5 A cutoff, 52 neighbors), ASE calculator, float32. Time = median of 3 calls, each three model passes. Peak mem = peak allocated GPU memory. Max |F| should be ~0 for the perfect crystal (checks the forces).
Red row: the first size that ran out of GPU memory. atomgptlab job 13474, 2026-09-27.

ALIGNN-FF results, energy only, H200 (1/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
1×1×1	4	0.5613	0.0668	0.6281	0.0	0.604015
2×2×2	32	0.0024	0.0023	0.0046	0.0	0.604017
3×3×3	108	0.0074	0.0025	0.0099	0.1	0.604017
4×4×4	256	0.0051	0.0025	0.0076	0.1	0.604020
5×5×5	500	0.0180	0.0032	0.0211	0.2	0.604019
6×6×6	864	0.0298	0.0035	0.0333	0.3	0.604021
7×7×7	1,372	0.0316	0.0049	0.0365	0.4	0.604014
8×8×8	2,048	0.0382	0.0067	0.0449	0.6	0.604023
9×9×9	2,916	0.0411	0.0093	0.0504	0.8	0.604024
10×10×10	4,000	0.0441	0.0121	0.0562	1.0	0.604011
11×11×11	5,324	0.0619	0.0157	0.0776	1.4	0.604010
12×12×12	6,912	0.0645	0.0203	0.0848	1.8	0.604036
13×13×13	8,788	0.0721	0.0254	0.0976	2.3	0.604000
14×14×14	10,976	0.0987	0.0313	0.1301	2.8	0.603929
15×15×15	13,500	0.1098	0.0382	0.1480	3.5	0.603846
16×16×16	16,384	0.1138	0.0460	0.1598	4.2	0.603869
17×17×17	19,652	0.1256	0.0561	0.1818	5.0	0.603917
18×18×18	23,328	0.1608	0.0648	0.2257	5.9	0.603956
19×19×19	27,436	0.1737	0.0762	0.2499	7.0	0.603955
20×20×20	32,000	0.1851	0.0884	0.2735	8.1	0.603858
21×21×21	37,044	0.2089	0.1022	0.3110	9.4	0.603743
22×22×22	42,592	0.2497	0.1174	0.3671	10.8	0.603744
23×23×23	48,668	0.2379	0.1339	0.3718	12.4	0.603927
24×24×24	55,296	0.2551	0.1519	0.4070	14.0	0.604197
25×25×25	62,500	0.2764	0.1720	0.4484	15.9	0.604641
26×26×26	70,304	0.3377	0.1929	0.5306	17.8	0.605063
27×27×27	78,732	0.3614	0.2159	0.5773	20.0	0.605456
28×28×28	87,808	0.3882	0.2408	0.6291	22.3	0.605646
29×29×29	97,556	0.5792	0.2674	0.8466	24.8	0.605734
30×30×30	108,000	0.6158	0.2960	0.9118	27.4	0.605611

B200 settings rerun on one NVIDIA H200 NVL: Cu FCC, 5 A cutoff, 12 neighbors, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (matches the B200 DGL energies within 1e-5 eV/atom up to 500 atoms). One call per size; the first row i GPU warm-up. Largest size shown is the largest that completed; the next (53×53×53 = 595,508 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600).
Timings are unaffected. Skipjack job 920089, 2026-09-28.

ALIGNN-FF results, energy only, H200 (2/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
31×31×31	119,164	0.6895	0.3260	1.0156	30.2	0.604993
32×32×32	131,072	0.7279	0.3587	1.0866	33.2	0.604449
33×33×33	143,748	0.8287	0.3946	1.2233	36.5	0.603987
34×34×34	157,216	0.8357	0.4293	1.2650	39.9	0.603673
35×35×35	171,500	0.8934	0.4685	1.3619	43.5	0.602742
36×36×36	186,624	1.0349	0.5095	1.5444	47.3	0.602057
37×37×37	202,612	1.1004	0.5565	1.6569	51.4	0.601320
38×38×38	219,488	1.2563	0.5999	1.8562	55.6	0.601225
39×39×39	237,276	1.3330	0.6514	1.9845	60.1	0.600928
40×40×40	256,000	1.5010	0.7002	2.2011	64.9	0.600853
41×41×41	275,684	1.5399	0.7535	2.2934	69.9	0.600795
42×42×42	296,352	1.5578	0.8105	2.3683	75.1	0.600620
43×43×43	318,028	1.7216	0.8683	2.5899	80.6	0.600554
44×44×44	340,736	1.8298	0.9311	2.7609	86.4	0.600510
45×45×45	364,500	2.0288	1.0003	3.0292	92.4	0.600274
46×46×46	389,344	2.1526	1.0680	3.2206	98.7	0.600081
47×47×47	415,292	2.3286	1.1412	3.4699	105.3	0.599898
48×48×48	442,368	2.4385	1.2090	3.6476	112.1	0.600148
49×49×49	470,596	2.4171	1.4137	3.8308	119.3	0.602111
50×50×50	500,000	2.5177	1.5239	4.0417	126.7	0.619691
51×51×51	530,604	2.7645	1.6988	4.4633	134.5	0.639844
52×52×52	562,432	2.9893	1.9614	4.9507	142.5	0.658485

B200 settings rerun on one NVIDIA H200 NVL: Cu FCC, 5 A cutoff, 12 neighbors, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (matches the B200 DGL energies within 1e-5 eV/atom up to 500 atoms). One call per size; the first row i GPU warm-up. Largest size shown is the largest that completed; the next (53×53×53 = 595,508 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600).
Timings are unaffected. Skipjack job 920089, 2026-09-28.

ALIGNN-FF results with forces, H200

Supercell	Atoms	Time (s)	Peak mem (GB)	MB/atom	Energy (eV/atom)	Max |F| (eV/A)
2×2×2	64	0.020	0.2	3.05	-8.7476	2.2e-05
4×4×4	512	0.032	1.1	2.09	-8.7476	3.6e-05
6×6×6	1,728	0.075	3.4	1.99	-8.7476	3.7e-05
8×8×8	4,096	0.159	8.1	1.97	-8.7475	3.9e-05
10×10×10	8,000	0.278	15.7	1.96	-8.7479	3.8e-05
12×12×12	13,824	0.470	27.0	1.96	-8.7479	4.0e-05
14×14×14	21,952	0.724	42.9	1.95	-8.7476	5.6e-05
15×15×15	27,000	0.905	52.7	1.95	-8.7470	5.8e-05
16×16×16	32,768	1.175	64.0	1.95	-8.7464	5.6e-05
17×17×17	39,304	1.339	76.8	1.95	-8.7467	5.7e-05
18×18×18	46,656	1.578	91.1	1.95	-8.7470	5.7e-05
19×19×19	54,872	1.887	107.1	1.95	-8.7475	5.6e-05
20×20×20	64,000	2.184	124.9	1.95	-8.7482	5.8e-05
21×21×21	74,088	2.534	144.6	1.95	-8.7482	5.8e-05
22×22×22	85,184	OOM	–	–	–	–

Energy + forces + stress single point on one NVIDIA H200 NVL: Si diamond (8-atom conventional cell, i×i×i), then-default matpes_r2scan checkpoint (hidden 128, 2+2 layers, 5 A cutoff, 52 neighbors), ASE calculator, float32. Time = median of 3 calls, each three model passes. Peak mem = peak allocated GPU memory. Max |F| should be ~0 for the perfect crystal (checks the forces).
Red row: the first size that ran out of GPU memory. Skipjack job 913733, 2026-09-27.

ALIGNN-FF largest single evaluation, GB10

Model	Path	Atoms	Peak mem (GB)	MB/atom	Time (s)	Max dE (eV/atom)
AFF-r2SCAN	energy only	551,368	107.3	0.195	9.7	4.4e-06
AFF-r2SCAN	E+F+S, ASE	54,872	107.2	1.953	7.6	1.2e-03
AFF-r2SCAN	E+F+S, tensors	140,608	105.3	0.749	4.6	1.1e-05
AFF-r2SCAN	E+F+S, tensors + ckpt	195,112	102.5	0.526	8.5	1.1e-05
alignn_ff_mptrj	energy only	238,328	92.0	0.386	13.1	2.6e-06
alignn_ff_mptrj	E+F+S, ASE	13,824	97.0	7.019	6.5	1.9e-04
alignn_ff_mptrj	E+F+S, tensors	39,304	97.5	2.481	4.7	2.1e-06
alignn_ff_mptrj	E+F+S, tensors + ckpt	97,336	110.9	1.139	16.0	2.5e-06
alignn_ff_matpes_pbe	energy only	238,328	92.0	0.386	13.1	9.7e-07
alignn_ff_matpes_pbe	E+F+S, ASE	13,824	97.0	7.019	6.5	2.0e-04
alignn_ff_matpes_pbe	E+F+S, tensors	39,304	97.5	2.481	4.8	2.0e-06
alignn_ff_matpes_pbe	E+F+S, tensors + ckpt	97,336	110.9	1.139	16.0	2.0e-06
alignn_ff_matpes_r2scan	energy only	262,144	101.2	0.386	15.0	2.2e-06
alignn_ff_matpes_r2scan	E+F+S, ASE	13,824	97.0	7.019	6.5	1.3e-04
alignn_ff_matpes_r2scan	E+F+S, tensors	46,656	115.7	2.481	5.6	6.6e-06
alignn_ff_matpes_r2scan	E+F+S, tensors + ckpt	97,336	110.9	1.139	15.9	7.8e-06

Largest pristine Si cell (8 i^3 atoms) evaluated on one GB10, one process per path (2026-11-01,
jobs 19126-19129). The next size was killed by the host (exit 137). Time = median of 3 calls
(through ASE, each call is three model passes). Tensors = forward_tensors (the LAMMPS path, no second-order graph).
Ckpt = layer checkpointing. Max dE = largest change in energy per atom from the small-cell value over the sweep.
Not comparable to the Cu energy-only runs (older model, 12 neighbors).

ALIGNN-FF chunked runs, GB10

Model	Atoms	GB10s	Computes	Cubes	Largest piece	Peak mem (GB)	Time (s)	E (eV/atom)
AFF-r2SCAN	8M	1	energy	64	261,205	49.3	264	-8.69961
AFF-r2SCAN	8M	1	forces	216	106,067	45.0	979	-8.69961
AFF-r2SCAN	64M	4	energy	512	261,206	49.3	351	-8.69959
AFF-r2SCAN	64M	5	forces	1,728	106,054	45.0	1,379	-8.69959
AFF-r2SCAN	1B	3	energy	8,000	261,222	49.3	3,235	-8.69960
AFF-r2SCAN	1B	4	forces	46,656	71,579	30.2	14,522	-8.69960
alignn_ff_mptrj	8M	1	energy	216	106,002	39.3	1,178	-5.35097
alignn_ff_mptrj	8M	1	forces	512	57,375	49.9	4,620	-5.35097
alignn_ff_mptrj	64M	4	energy	1,331	128,175	47.6	1,473	-5.35095
alignn_ff_mptrj	64M	6	forces	4,096	57,379	49.9	6,231	-5.35095
alignn_ff_matpes_pbe	8M	1	energy	216	106,002	39.3	1,176	-5.38137
alignn_ff_matpes_pbe	8M	1	forces	512	57,375	49.9	4,621	-5.38137
alignn_ff_matpes_r2scan	8M	1	energy	216	106,002	39.3	1,175	-8.72789
alignn_ff_matpes_r2scan	8M	1	forces	512	57,375	49.9	4,608	-8.72789

Rattled Si (0.05 A), energy and forces, no stress (2026-11-01 to 03, jobs 19140-19222). Each cube is evaluated with a halo on one GPU. Force runs use layer checkpointing. Time = slowest task. The AFF-r2SCAN 64M force run was 6 tasks on 5 nodes (one node ran two in turn, 2,689 s), and the 64M energy runs 6 tasks on 4. The 1B runs were 10 tasks, on 4 nodes with forces (13 h from first start to last finish) and 3 without. Peak mem = GPU memory of the largest piece. Every task also holds the whole cell on the host, up to 73.9 GB per task at 1B. These are the largest sizes tried: none reached a memory limit. AFF-r2SCAN: exact 20.01 A halo. alignn_ff_* models: 20 A halo instead of the exact 40 A, which changed forces by 2e-9 to 2e-8 eV/A on 8,000 atoms of Si and 1.2e-8 to 1.3e-7 on SiGe in float64. In float32, chunked forces differ from the full cell by 1.4e-4 to 9.9e-4 eV/A even at the exact halo, which is roundoff (2e-12 eV/A in float64). Tasks started as nodes freed up, so the 64M force runs took 45 min (AFF-r2SCAN) and 1 h 59 min (alignn_ff_mptrj) from first start to last finish.

VASP results

Supercell	Atoms	GPUs	k-points	Bands	SCF cycles	Elapsed (s)	s / SCF	E (eV/atom)	Status
3×3×3	54	1	3	136	20/20	15.3	0.76	-5.4105	converged
3×3×3	54	2	3	136	20/20	17.8	0.89	-5.4105	converged
3×3×3	54	4	3	136	20/20	18.3	0.91	-5.4105	converged
3×3×3	54	8	3	136	20/20	27.3	1.36	-5.4105	converged
4×4×4	128	1	3	321	20/20	39.6	1.98	-5.4211	converged
4×4×4	128	2	3	321	20/20	38.3	1.92	-5.4211	converged
4×4×4	128	4	3	321	20/20	36.5	1.82	-5.4211	converged
4×4×4	128	8	3	321	20/20	42.5	2.13	-5.4211	converged
5×5×5	250	1	1	626	20/20	57.6	2.88	-5.3903	converged
5×5×5	250	2	1	626	20/20	62.4	3.12	-5.3903	converged
5×5×5	250	4	1	628	20/20	75.3	3.76	-5.3903	converged
5×5×5	250	8	1	632	20/20	108.1	5.41	-5.3903	converged
6×6×6	432	1	1	1,081	20/20	113.7	5.68	-5.4111	converged
6×6×6	432	2	1	1,082	20/20	96.2	4.81	-5.4111	converged
6×6×6	432	4	1	1,084	20/20	89.3	4.46	-5.4111	converged
6×6×6	432	8	1	1,088	20/20	123.6	6.18	-5.4111	converged
10×10×10	2,000	4	1	5,004	5/5	518.4	103.69	-5.5124	not converged
12×12×12	3,456	8	1	8,648	2/2	678.6	339.29	-3.1397	not converged
14×14×14	5,488	8	1	13,728	2/2	1561.4	780.69	-3.1433	not converged
15×15×15	6,750	8	1	16,880	0/2	–	–	–	crashed
16×16×16	8,192	8	1	20,488	0/2	–	–	–	crashed

VASP GPU build, single-point SCF. 3×3×3 to 6×6×6: NELM=20, EDIFF=1E-6. 10×10×10: NELM=5, ALGO=Fast.
12×12×12 and larger: NELM=2, ALGO=Fast. SCF cycles = completed / maximum allowed (NELM).
E = energy of the last SCF cycle / atom count. Converged = last energy change below the run's EDIFF.
Amber rows were stopped early for timing; their E is an unconverged intermediate value, not a result.
Red rows crashed before the first SCF cycle (MPI segfault, not out of memory).